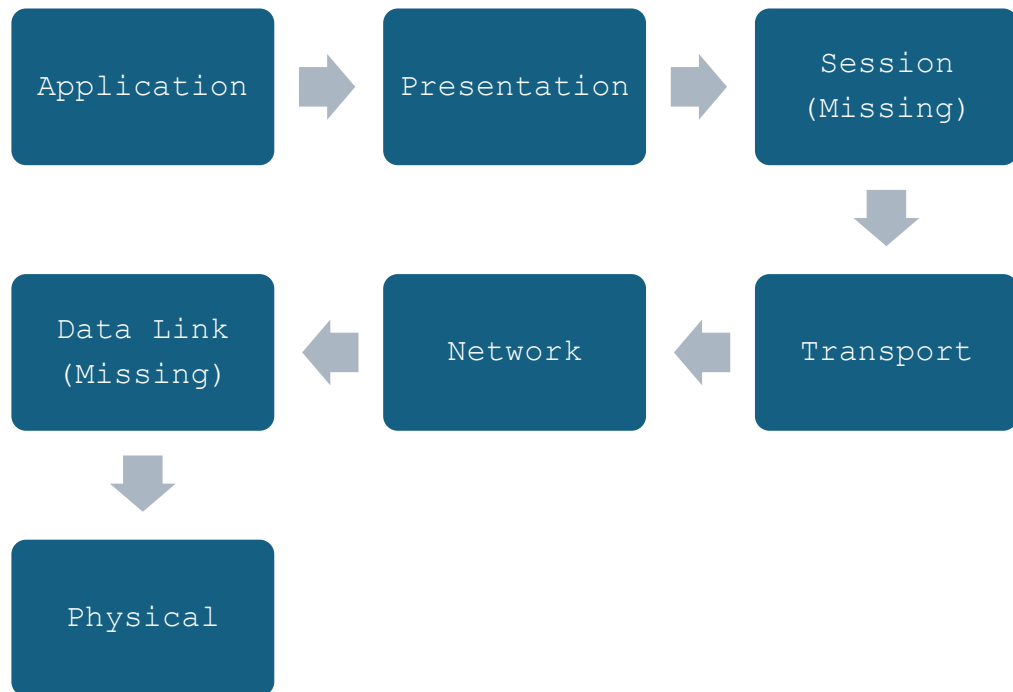


1.Missing OSI Layers



- The missing layers are session layer and Data Link layer
- Session Layer:
 - Establishes, manages, synchronizes and terminates communication sessions.
 - The session layer maintains continuous communication between sender and receiver.
- Data link layer:
 - Performs framing, MAC addressing, error detection and flow control.
 - The data link layer ensures reliable node to node transmission by detecting transmission errors.
- Together with the remaining OSI layers, they provide secure and reliable communication.

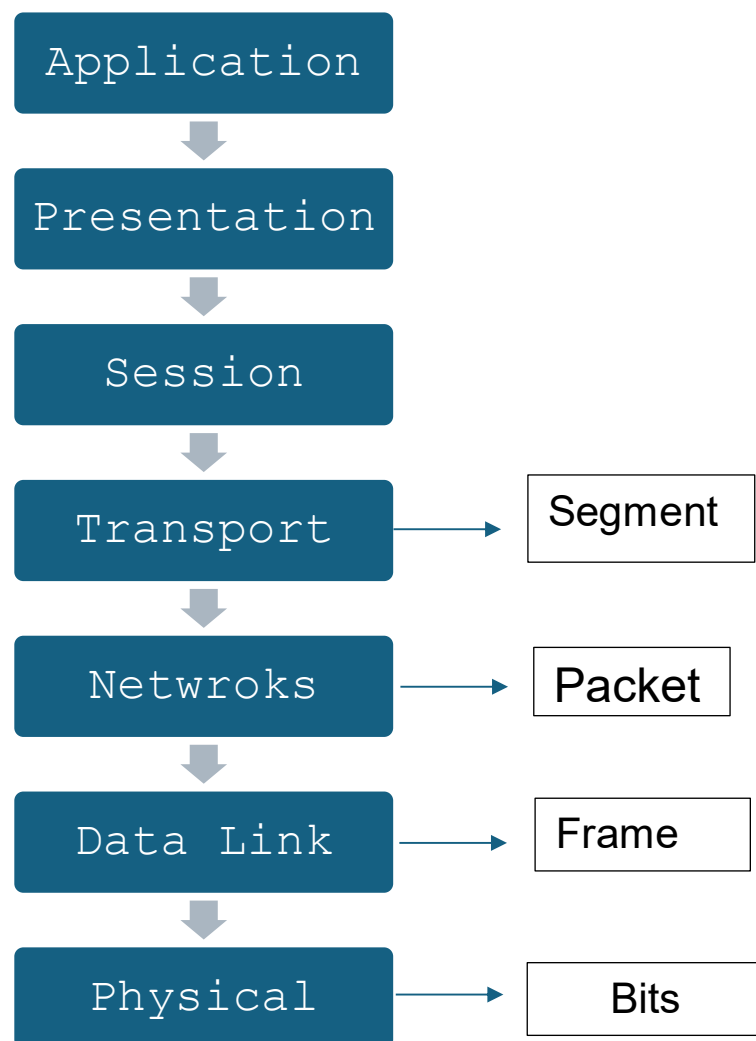
Application

Example: While accessing a website, the Session Layer keeps the browsing session active, and the Data Link Layer ensures error-free transmission over Wi-Fi or Ethernet.

Conclusion

Every OSI layer has a specific responsibility. The Session Layer manages communication sessions, while the Data Link Layer ensures reliable local delivery, making data transmission efficient.

2.Data units and OSI layers



- Data moves through the OSI layers using **encapsulation**.
- The Transport Layer divides data into **Segments**.
- The Network Layer adds IP information to create **Packets**.
- The Data Link Layer adds MAC addresses to create **Frames**.
- The Physical Layer converts frames into **Bits** for transmission.
- At the receiver, **decapsulation** reconstructs the original message.

Application

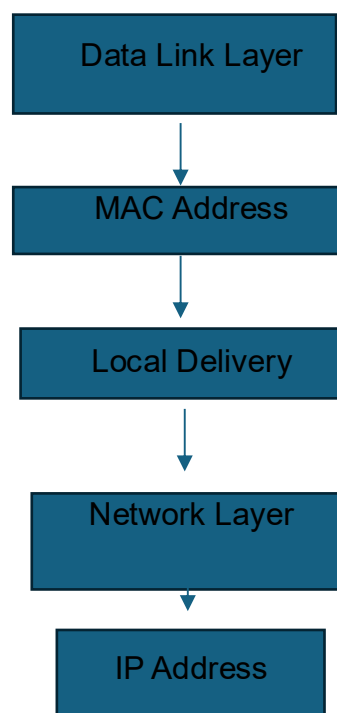
When sending an email, the message is converted from:

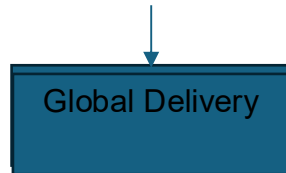
Segments → Packets → Frames → Bits before transmission.

Conclusion

The transformation of data units ensures reliable communication, proper addressing, routing, and error detection.

Q3. MAC Address and IP Address





- MAC Address uniquely identifies a device within a local network.
- IP Address identifies a device across different networks.
- Switches use MAC addresses for local forwarding.
- Routers use IP addresses for routing between networks.
- Local and global addressing work together for complete communication.
- Both addresses are essential for accurate data delivery.

Application

When a user accesses a website, routers use IP addresses to reach the destination network, while switches use MAC addresses to deliver data to the correct device.

Conclusion

MAC addresses ensure local delivery, and IP addresses ensure global delivery, enabling successful end-to-end communication.

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